Reading 1

Reading & understanding

Words in use

Language focus

Complete the sentences with the words below. Change the form if necessary.

counteract	erosion	finite	metabolic
prolong	skeptical	thermal	unprecedented

1. Research suggests that our planet is warming at a(n) unprecedented pace.
2. Your metabolic rate is the speed at which your body transforms food into energy.
3. The research team proposes a few key recommendations to counteract the declining of domestic demand.
4. A(n) thermal management system optimizes battery range in high and low temperatures, allowing the car to maintain optimal range capability.

Reading 1

Reading & understanding

Words in use

Language focus

Complete the sentences with the words below. Change the form if necessary.

counteract	erosion	finite	metabolic
prolong	skeptical	thermal	unprecedented

5. By the 1920s, ___erosion___ from winds and weather had removed a large part of the land.
6. It's needless to say that the value of water, this ___finite___ and irreplaceable resource, is enormous.
7. Experts say the spread of the virus in countries with little protection from vaccines could lead to more variants and ___prolong___ the pandemic.
8. Many people are still ___skeptical___ about artificial intelligence getting adopted by financial institutions.

Reading 1

Reading & understanding

Translation

Language focus

Translate the following sentences into Chinese. Pay attention to the words in bold.

1. A landmark new report on biodiversity and ecosystem was released in Paris on Monday, warning about the **unprecedented** declining rate of nature globally.

   周一，一份关于生物多样性和生态系统的里程碑式的新报告在巴黎发布，对全球自然界前所未有的衰退速度发出了警告。

2. Soil **erosion**—the loss of fertile topsoil by water, wind, or tillage (耕作)—is a global problem, the consequences of which are becoming more widely noticed.

   土壤侵蚀是水、风以及耕作导致的肥沃表土流失，这已是全球性问题，其后果正受到越来越广泛的关注。

Reading 1

Reading & understanding

Translation

Language focus

Translate the following sentences into Chinese. Pay attention to the words in bold.

3. If the right amount of the boat is submerged, the force of buoyancy (浮力) is strong enough to **counteract** gravity pulling the boat down, and the boat floats.

如果适量的船体没入水中，强大的浮力足以抵消将船向下拉坠的重力，船就会漂浮起来。

4. **The tropics** receive a great amount of direct solar energy, which produces more evaporation than higher latitudes.

热带地区接收到大量太阳直射的能量，比高纬度地区产生更多的蒸发。

Reading 1

Reading & understanding

Difficult sentences

Language points

01

Climate change 101 (Title)

在美国大学课程编号系统中，101通常用于标记某一领域或主题入门级的基础课程。本文作者运用101代表文章中所包含的信息是理解气候变化这一主题所需的最基本信息。

Reading 1

Reading & understanding

Difficult sentences

Language points

03

Eighty percent of this additional heat is absorbed by the world's oceans, the surface of which has increased in temperature by 0.13°F per decade since 1901 (NASA, n.d.). **(Para. 2)**

n.d.为no date的缩写。在学术写作中若年代不明，常用n.d.标注。

Reading 1

Reading & understanding

Difficult sentences

Language points

02

Since global temperature records began in the late 19th century, the planet's average surface temperature has risen by approximately 2.0°F (1.1°C), with 16 of the 17 hottest years on record occurring over the last 35 years. **(Para. 2)**

此句结构比较复杂，since引导时间状语从句，表示“自……以来”；主句使用现在完成时has risen，与since引导的表示一段时间的状语连用，指地表温度上升这一动作或状态从过去某时开始，持续到现在，还可能持续下去；主句之后是with引导的伴随状语，现在分词occurring强调occur这一动作或状态与主句状态同时进行。

Reading 1

Reading & understanding

Difficult sentences

Language points

04

Ice ages—driven by slow variations in the Earth's orbit that alter the distribution of the Sun's energy—have occurred, on average, every 100,000 years. (Para. 3)

此句主句为Ice ages have occurred every 100,000 years。driven引导的过去分词短语作后置定语，相当于定语从句which were driven …，修饰Ice ages，解释说明其形成的原因。句中破折号主要表示停顿，目的是把driven引导的插入成分隔开。

Reading 1

Reading & understanding

Difficult sentences

Language points

05

A blanket of greenhouse gases—carbon dioxide, water vapor, nitrous oxide, and methane—in the Earth's atmosphere absorb heat and warm the planet's surface, a natural phenomenon known as the greenhouse effect. (Para. 4)

此句中破折号的作用是用来解释名词短语A blanket of greenhouse gases的。本句后半部分为known引导的过去分词短语作后置定语，修饰名词短语a natural phenomenon。

Reading 1

Reading & understanding

Difficult sentences

Language points

06

Since the early 1800s, ocean acidity has increased by around 25%, depressing the metabolic rates of some species, lowering the immune responses of others, destabilizing marine ecosystems, and causing the bleaching, and possibly the eventual death, of the world's coral reefs. **(Para. 5)**

此句中，主句使用现在完成时表示动作或状态从过去某时开始，持续到现在，还可能持续下去；主句后四个现在分词短语并列，在句中作结果状语，说明海水酸度上升对海洋物种的影响，此处并列短语的使用有加强语气的作用。

Reading 1

Reading & understanding

Difficult sentences

Language points

07

Although the recent resolution reached in Paris is hugely significant, many claim it is too little, too late. **(Para. 8)**

此句中“在巴黎达成的决议”是指《巴黎协定》（Paris Agreement，全称Paris Agreement Under the United Nations Framework Convention on Climate Change），旨在大幅减少全球温室气体排放，将全球气温升幅限制在比工业化前水平高2℃以内，并寻求将气温升幅进一步限制在1.5℃以内的措施。